

Stores geographic and general information about airports. Attributes include `airport_id` (PK, Integer) as a unique identifier of the airport, `airport_name` (VarChar) for the full name of the airport, `country`, `state`, `city` (VarChar) for airport location details, and `created_at`, `updated_at` (Timestamp) for record metadata.

2. airline

Contains details about commercial air carriers operating at the airport. Attributes include `airline_id` (PK, Integer) as a unique identifier of the airline, `airline_code` (VarChar) for IATA/ICAO short code, `name` (VarChar) for the official name of the airline, `country` (VarChar) for the country of registration, and `created_at`, `updated_at` (Timestamp) for record metadata.

3. flight

Tracks scheduled and actual operational flights. Attributes include `flight_id` (PK, Integer) as a unique flight instance ID, `airline_id` (FK, Integer) for the operating airline, `departure_airport_id` (FK, Integer) for the origin airport, `arrival_airport_id` (FK, Integer) for the destination airport, `departing_gate`, `arriving_gate` (VarChar) for terminal gates, `scheduled_departure_time`, `scheduled_arrival_time` (Timestamp) for scheduled timetable, `actual_departure_time`, `actual_arrival_time` (Timestamp) for actual flight times, and `created_at`, `updated_at` (Timestamp) for record metadata.

4. passenger

Stores personal and identity verification details of passengers. Attributes include `passenger_id` (PK, Integer) as a unique passenger profile ID, `first_name`, `last_name` (VarChar) for the passenger name, `gender` (VarChar) for gender identification, `date_of_birth` (Date) for birth date, `country_of_citizenship`, `country_of_residence` (VarChar) for nationality details, `passport_number` (VarChar) for identification document number, and `created_at`, `updated_at` (Timestamp) for record metadata.

5. booking

Records passenger flight reservations and payment transactions. Attributes include `booking_id` (PK, Integer) as a unique reservation ID, `flight_id` (FK, Integer) for the associated flight, `passenger_id` (FK, Integer) for the associated passenger, `status` (VarChar) for booking status (e.g., Confirmed, Cancelled), `booking_platform` (VarChar) for booking source/website, `ticket_price` (Numeric) for price of the ticket, and `created_at`, `updated_at` (Timestamp) for record metadata.

6. booking_change

Tracks historical changes and modifications made to a booking. Attributes include `change_id` (PK, Integer) as a unique change record ID, `booking_id` (FK, Integer) for target booking reference, `change_details` (Text) for details regarding the modification, and `created_at`, `updated_at` (Timestamp) for record metadata.

7. boarding_pass

Details boarding passes issued to passengers after check-in. Attributes include **boarding_pass_id** (PK, Integer) as a unique boarding pass ID, **booking_id** (FK, Integer) for the associated booking, **seat** (VarChar) for assigned seat number, **boarding_time** (Timestamp) for time of boarding, and **created_at**, **updated_at** (Timestamp) for record metadata.

8. baggage

Registers registered luggage items linked to passenger bookings. Attributes include **baggage_id** (PK, Integer) as a unique luggage tag ID, **booking_id** (FK, Integer) for the associated booking, **weight_kg** (Numeric) for weight in kilograms, and **created_at**, **updated_at** (Timestamp) for record metadata.

9. baggage_check

Logs baggage screening and inspection outcomes at the airport. Attributes include **baggage_check_id** (PK, Integer) as a unique check ID, **baggage_id** (FK, Integer) for checked luggage, **passenger_id** (FK, Integer) for luggage owner, **check_results** (VarChar) for inspection status (e.g., Passed, Flagged), and **created_at**, **updated_at** (Timestamp) for record metadata.

10. security_check

Logs individual passenger security control checkpoints. Attributes include **security_check_id** (PK, Integer) as a unique security log ID, **passenger_id** (FK, Integer) for the screened passenger, **check_results** (VarChar) for security check outcome, and **created_at**, **updated_at** (Timestamp) for record metadata.

4. Relationships and Cardinality Constraints

The relationship from **airline** to **flight** (1:N) represents that one airline operates multiple scheduled flights. The relationship from **airport** to **flight** (1:N) represents that an airport serves as the departure/arrival point for many flights. The relationship from **flight** to **booking** (1:N) represents that one flight can have multiple passenger bookings. The relationship from **passenger** to **booking** (1:N) represents that one passenger can make multiple flight bookings. The relationship from **booking** to **booking_change** (1:N) represents that a single booking can undergo several changes. The relationship from **booking** to **boarding_pass** (1:1 / 1:N) represents that a valid booking leads to issuing a boarding pass. The relationship from **booking** to **baggage** (1:N) represents that one booking can include multiple registered pieces of baggage. The relationship from **baggage** to **baggage_check** (1:N) represents that each baggage item undergoes airport baggage checks. The relationship from **passenger** to **baggage_check** (1:N) links passenger ownership to baggage check logs. The relationship from **passenger** to **security_check** (1:N) represents that a passenger passes through airport security checks.